

Ex. No. 6	PERFORM SCHEMA DESIGN AND INDEXING
Date: 23:02:2026	

Aim

To perform Schema Design and Indexing in MongoDB.

Schema Design:

Let's assume a Digital Vehicle Maintenance & Tracking Database. We'll create a "vehicles" collection with fields like:

- `_id`
- `vehicleName`
- `ownerName`
- `serviceType`
- `serviceCount`
- `maintenanceStatus`

Data Insertion:

```
db.vehicles.insertMany([
```

```
{
  _id: 1,
  vehicleName: "Honda City",
  ownerName: "Mohammed Dhariq",
  serviceType: "Oil Change", serviceCost:
  3000,
  maintenanceStatus: "Completed"
},
```

```
{
  _id: 2,
  vehicleName: "Hyundai i20",
  ownerName: "Rahul", serviceType:
  "Brake Service", serviceCost:
  5000,
  maintenanceStatus: "Pending"
},
```

```
{
  _id: 3,
  vehicleName: "Toyota Innova",
  ownerName: "Arun", serviceType:
  "Engine Check", serviceCost:
  7000,
  maintenanceStatus: "Completed"
},
```

```
{
  _id: 4,
  vehicleName: "Kia Seltos",
  ownerName: "Karthik", serviceType:
  "Battery Service", serviceCost:
  4000, maintenanceStatus: "In
  Progress"
},
```

```
{
  _id: 5,
```

```
vehicleName: "Mahindra XUV700", ownerName:
"Vignesh", serviceType: "Full Service",
serviceCost: 9000, maintenanceStatus:
"Completed"
}

]);
```

Indexing:

1. Create index on the serviceType field to speed up service-based queries: Query

```
> db.vehicles.createIndex({ serviceType: 1 });
```

2. Create compound index serviceCost

and maintenanceStatus:

: Query

```
> db.vehicles.createIndex({ serviceCost: 1, maintenanceStatus: -1 });
```

Queries:

Find all vehicles with "Oil Change" service type:

```
> db.vehicles.find({ serviceType: "Oil Change" });
```

Output:

```
[
  {
    "_id": 1,
    "vehicleName": "Honda City",
    "ownerName": "Mohammed Dhariq",
    "serviceType": "Oil Change",
    "serviceCost": 3000,
    "maintenanceStatus": "Completed"
  }
]
```

Find vehicles with service cost less than 5000:

```
> db.vehicles.find({ serviceCost: { $lt: 5000 } });
```

Output:

```
[
  {
    "_id": 1,
    "vehicleName": "Honda City",
    "ownerName": "Mohammed Dhariq",
    "serviceType": "Oil Change",
    "serviceCost": 3000,
    "maintenanceStatus": "Completed"
  },
  {
    "_id": 4,
    "vehicleName": "Kia Seltos",
    "ownerName": "Karthik",
    "serviceType": "Battery Service",
    "serviceCost": 4000,
    "maintenanceStatus": "In Progress"
  }
]
```

Check all created indexes:

```
> db.vehicles.getIndexes();
```

Output:

```
[
  {
    "v": 2,
    "key": { "_id": 1 },
    "name": "_id_"
  },
  {
    "v": 2,
    "key": { "serviceType": 1 },
    "name": "serviceType_1"
  },
]
```

```
{  
  "v": 2,  
  "key": { "serviceCost": 1, "maintenanceStatus": -1 },  
  "name": "serviceCost_1_maintenanceStatus_-1"  
}  
]
```

Result:

Thus, the schema design and indexing operations on vehicle maintenance collection in MongoDB were successfully performed.